

Proof. **Step1. Degenerate or edge cases.** If $n = 1$ then both sides of

$$n \sum_{i=1}^n a_i b_i = a_1 b_1 \quad \text{and} \quad \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) = a_1 b_1$$

coincide, so the inequality holds with equality.

If the a -sequence is constant, say $a_i = a$ for all i , then

$$n \sum_{i=1}^n a_i b_i = na \sum_{i=1}^n b_i = \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right),$$

and similarly when the b -sequence is constant. Hence the statement is true in all degenerate situations.

Step2. Centered variables and means. Define the arithmetic means

$$\bar{a} := \frac{1}{n} \sum_{i=1}^n a_i, \quad \bar{b} := \frac{1}{n} \sum_{i=1}^n b_i,$$

and the centred sequences

$$\tilde{a}_i = a_i - \bar{a}, \quad \tilde{b}_i = b_i - \bar{b} \quad (i = 1, \dots, n).$$

Subtracting a constant does not affect the ordering, so

$$\tilde{a}_1 \leq \tilde{a}_2 \leq \dots \leq \tilde{a}_n, \quad \tilde{b}_1 \leq \tilde{b}_2 \leq \dots \leq \tilde{b}_n. \quad (2)$$

Moreover,

$$\sum_{i=1}^n \tilde{a}_i = 0, \quad \sum_{i=1}^n \tilde{b}_i = 0. \quad (3)$$

Step3. Non-negativity of the covariance (Rearrangement Inequality). For two sequences sorted in the same non-decreasing order, the Rearrangement Inequality states that for any permutation σ of $\{1, \dots, n\}$

$$\sum_{i=1}^n x_i y_{\sigma(i)} \leq \sum_{i=1}^n x_i y_i,$$

with equality only when the sequences are linearly dependent after centring. Apply this to $(\tilde{a}_i)$ and $(\tilde{b}_i)$. Because of (2) they are similarly ordered; for the reverse permutation $\rho(i) = n + 1 - i$ we obtain

$$\sum_{i=1}^n \tilde{a}_i \tilde{b}_{\rho(i)} \leq \sum_{i=1}^n \tilde{a}_i \tilde{b}_i. \quad (4)$$

Using (3),

$$\sum_{i=1}^n \tilde{a}_i \tilde{b}_{\rho(i)} = \sum_{i=1}^n \tilde{a}_i (\tilde{b}_{n+1-i}) = - \sum_{i=1}^n \tilde{a}_i \tilde{b}_i,$$

so (4) becomes

$$-\sum_{i=1}^n \tilde{a}_i \tilde{b}_i \leq \sum_{i=1}^n \tilde{a}_i \tilde{b}_i,$$

hence

$$\boxed{\sum_{i=1}^n \tilde{a}_i \tilde{b}_i \geq 0}. \quad (5)$$

Thus the covariance of two similarly ordered centred sequences is non-negative.

Step4. Algebraic manipulation to Chebyshev's form. Expand each product in (5):

$$\tilde{a}_i \tilde{b}_i = (a_i - \bar{a})(b_i - \bar{b}) = a_i b_i - a_i \bar{b} - \bar{a} b_i + \bar{a} \bar{b}.$$

Summing over i and using linearity,

$$\sum_{i=1}^n \tilde{a}_i \tilde{b}_i = \sum_{i=1}^n a_i b_i - \bar{b} \sum_{i=1}^n a_i - \bar{a} \sum_{i=1}^n b_i + n \bar{a} \bar{b}.$$

Since $\sum_{i=1}^n a_i = n \bar{a}$ and $\sum_{i=1}^n b_i = n \bar{b}$,

$$\sum_{i=1}^n \tilde{a}_i \tilde{b}_i = \sum_{i=1}^n a_i b_i - \frac{(\sum_{i=1}^n a_i)(\sum_{i=1}^n b_i)}{n}.$$

Insert this expression into (5) to obtain

$$\sum_{i=1}^n a_i b_i \geq \frac{(\sum_{i=1}^n a_i)(\sum_{i=1}^n b_i)}{n}.$$

Multiplying by n yields Chebyshev's sum inequality

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right),$$

which is precisely the desired statement (1).

Step5. Equality conditions. Equality in (5) occurs exactly when equality holds in the Rearrangement Inequality for the centred sequences. The Rearrangement Inequality is strict unless there exists a constant λ such that

$$\tilde{a}_i = \lambda \tilde{b}_i \quad \text{for every } i. \quad (6)$$

Re-expressing (6) in the original variables gives

$$a_i = \lambda b_i + \mu, \quad \mu := \bar{a} - \lambda \bar{b}, \quad (7)$$

so equality holds in one of the following mutually exclusive situations:

1. All a_i are equal ($\tilde{a}_i = 0$ for all i);
2. All b_i are equal ($\tilde{b}_i = 0$ for all i);
3. There exist constants $\lambda \geq 0$ and μ such that $a_i = \lambda b_i + \mu$ for every i (an affine relation preserving the order).

These are exactly the equality cases for Chebyshev's sum inequality.

Step6. Conclusion. All degenerate cases have been handled, the centred covariance has been shown to be non-negative via the Rearrangement Inequality, the algebraic passage to Chebyshev's form is rigorous, and the equality conditions are completely characterised. Hence for any two non-decreasing real sequences $(a_i)_{i=1}^n$ and $(b_i)_{i=1}^n$ we have

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right),$$

with equality precisely in the cases listed in Step5. $\square$

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